a
Zero-shot generalization tests (all timelapse 3D)
MegaSeg training
WTC-11 hiPSC Single-Cell Image Dataset (Viana et al., Nature 2023)
Cells: live hiPSC (WTC-11)
Microscope: Zeiss spinning disk
Objective: 100×/1.25 W, 1.2× adapter
Voxel: 0.108 × 0.108 × 0.108 µm
Acquisition: single 3D stack per FOV
Structures: 13 proteins (endogenous mEGFP)
b Tubulin
c Mitochondria
d PCNA
e Cardio
f NPM1
Microscope
Airyscan 40x
Airyscan 40x
ZSD 100×
3iW1-0 63×/1.2 W
20/40/100× (multi)
Cell type
hiPSC
hiPSC
hiPSC
Cardiomyocyte
hiPSC
Protein
TUBA1B (new)
TOMM20 (seen)
PCNA (new)
Tnn + Titin (new)
NPM1 (new)
same as training
shifted vs training
b
Tubulin (Airyscan 40x, 20170606_C04_001 scene 0): Raw + MegaSeg mask (vol p5.0-p99.9, gamma=1.5, corrected 10 µm scale bar)
c
Mitochondria (Airyscan 40x, 20170424_C01_001 scene 1): Raw + MegaSeg mask (per-slice p5.0-p99.5, gamma=0.7, corrected 10 µm scale bar)
d
PCNA (ZSD, 20210518_F03_001 scene P36-D6): Raw + MegaSeg mask (per-slice p1.0-p99.0, 10 µm scale bar labeled)
e
Cardiomyocyte (20180327_C09_001): z=25 vs z=27 with colocalization (corrected 10 µm scale bar)
NPM1 across magnifications (3D: top / center / bottom; common 60 µm × 60 µm field)
ZSD 20x
Airyscan 40x
ZSD 100x (2×2 binned)